Exercise Sheet 3 (Lectures 5 and 6)

Short-Answer Questions

Exercise 1 (LU Decomposition):

Suppose an invertible matrix A has the decomposition $A = LU$. Which crucial properties do L and U have? Describe how to solve $Ax = b$ efficiently.

Exercise 2 (Transpose of a Matrix):

Write down the transpose A^T of

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \end{pmatrix}.$$

Questions on Lecture 5

Exercise 3 (Forward Substitution):

Solve the system

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 3 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \\ 8 \end{pmatrix}.$$

Explain why no Gaussian elimination is needed.

Exercise 4 (Constructing the L-Matrix):

Consider

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 3 \\ 1 & 7 & 7 \end{pmatrix}.$$

a) Use Gaussian elimination to put A into row echelon form U, without interchanging or scaling rows. Record each multiplier m_{ij} used in an operation

$$R_i \leftarrow R_i - m_{ij} R_j$$

in position (i, j) of a matrix L with diagonal entries 1 and zeros above the diagonal.

b) Compute LU and compare the result with A.

Exercise 5 (Matrices as Functions):

For $A \in \mathbb{R}^{m\times n}$, consider the function

$$f_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m, \qquad x \longmapsto Ax.$$

a) Interpret $\mathcal{C}(A)$ in terms of this function.
b) Suppose $\operatorname{rank}(A) = n$, or equivalently, $\mathcal{N}(A) = \{0\}$. What does this mean for f_A?

Exercise 6 (Polynomial Interpolation):

Let $(x_1, y_1), \ldots, (x_4, y_4)$ be four points with pairwise distinct x-coordinates. Describe how to find a polynomial of degree at most three,

$$p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3,$$

that passes through all four points. You may assume that such a polynomial exists.

Apply your method to

$$(0, 1), \qquad (1, 2), \qquad (2, 9), \qquad (3, 28).$$

Exercise 7 (Solving a System with Python):

Use Python and NumPy to solve

$$Ax = b, \qquad A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 5 & 0 \\ -1 & 1 & 10 \end{pmatrix}, \qquad b = \begin{pmatrix} 2 \\ 12 \\ 31 \end{pmatrix}.$$

Use `numpy.linalg.solve` and check your answer by calculating the residual $Ax - b$.

Questions on Lecture 6

Exercise 8 (Orthogonality and Linear Combinations):

Let $v, w_1, \ldots, w_n \in \mathbb{R}^m$. Show that if v is orthogonal to every w_j, then it is orthogonal to every vector in $\operatorname{span}\{w_1, \ldots, w_n\}$.

Exercise 9 (Null Space and Orthogonality):

Let

$$A = \begin{pmatrix} 1 & 2 & -1 & 0 \\ 2 & 5 & -1 & 3 \end{pmatrix}.$$

Find a basis of $\mathcal{N}(A)$ and calculate the scalar product of each basis vector with each row of A. Are the results surprising?

Exercise 10 (Column Scaling and Gram Matrix):

Consider

$$A = \begin{pmatrix} 1 & 10^8 \\ 0 & 10^8 \end{pmatrix}.$$

Compute the Gram matrix $A^T A$.

We want to use the scalar products in the Gram matrix to discover "correlations" between the columns of A. What could be a potential problem here and how can it be solved?

Exercise 11 (Transpose and Invertibility):

Show that $A \in \mathbb{R}^{n \times n}$ is invertible if and only if A^T is invertible.